

# JOY ROY

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## EDUCATION

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**Ph.D. Biomedical Informatics**, University of Pittsburgh School of Medicine 2020 - 2024

**B.A. Mathematics**, University of Maryland Baltimore County 2015 - 2019

## SKILLS

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**Technical** C/C++, Python, Bash, SQL, R, Matlab, LaTeX  
**Packages** Qt, Numpy, Pandas, Nipype, Scikit-learn, PyTorch, Seaborn, NetworkX, Nibabel, FSL  
**Other** Agile development, Object Oriented Programming, Docker, Linux/Unix, Adobe Suite, Jira/Confluence, Google Cloud Platform, Git, ANTS, Data Visualization, Scientific Writing  
**Certification** Interim DOD Secret Clearance

## EXPERIENCE

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**Senior Data Scientist** 2025 - Present  
Mayo Clinic *Remote - Baltimore, MD*

- Design, develop, and evaluate generative and agentic AI systems to improve healthcare delivery.
- Lead clinical research and validation studies, including IRB protocol development, study design, statistical analysis, and assessment of model performance, safety, and bias/fairness.
- Collaborate with physicians, multidisciplinary research teams, and industry partners (e.g., Microsoft) to implement and evaluate AI-driven clinical tools in real-world settings.

**Health Informatics Scientist** 2025 - 2025  
Henry M Jackson Foundation *Bethesda, MD*

- Designed a framework to harmonize heterogeneous research datasets into a centralized data lake, enabling downstream machine learning and analytics
- Led cross-disciplinary discussions with physicians, psychologists, and researchers to define requirements for data storage, integration, and longitudinal tracking
- Appointed to NICOE at Walter Reed Hospital with Interim DOD Secret Clearance

**Graduate Research Assistant** 2020 - 2024  
Pediatric Imaging Research Center, Children's Hospital of Pittsburgh *Pittsburgh, PA*

- Developed custom commandline software using Bash, Python, and Docker.
- Developed, trained, and evaluated machine learning models, including Random Forests and Convolutional Neural Networks, to derive predictive insights from large medical imaging datasets.
- Automated MRI preprocessing pipelines with custom Python scripts, utilizing libraries such as FSL and NumPy to perform tasks like segmentation, registration, and data normalization.
- Trained deep learning models on Google Cloud Compute Engine and Slurm-based HPC clusters.

**Software Engineer** 2019 - 2020  
Hughes Network Systems *Gaithersburg, MD*

- Designed and implemented real time software for protocols, algorithms, and products using C/C++
- Researched and tested system solutions to make applications perform optimally over IP-based wireless networks.
- Maintained Jenkins pipelines for automated unit testing and continuous integration and continuous development.